IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

DIGITAL IMAGE PROCESSING

Friday 12 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks

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1. Explain **qualitatively** what effect the following transformations will have on the Discrete Fourier Transform (DFT) of an image:

(i) Random rearrangement of the intensity values of the image pixels. **[4]**
(ii) Convolution of the image with a Gaussian kernel of minimal (almost zero) standard deviation. **[4]**
(iii) Addition of zero mean Gaussian noise to the image, independent of the image signal, with non-negligible standard deviation σ. **[4]**

(A qualitative explanation would describe the general behavior and characteristics of the effects of the transformations on the DFT, without necessarily providing specific numerical values or measurements.)

(b) Consider the population of vectors $\underline{f}$ of the form:

$$\underline{f}(x,y) = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ f_3(x,y) \end{bmatrix}$$

Each component $f_i(x,y),\ i = 1,2,3$ represents an image of size $M \times M$ where M is even. The population arises from the formation of the vectors $\underline{f}$ across the entire collection of pixels (x,y). The three images are defined as follows:

$$f_1(x,y) = \begin{cases} r_1 & 1 \le x \le \dfrac{M}{2}, 1 \le y \le M \\ r_2 & \dfrac{M}{2} < x \le M, 1 \le y \le M \end{cases}$$

$$f_2(x,y) = \begin{cases} r_3 & 1 \le y \le \dfrac{M}{2}, 1 \le x \le M \\ r_4 & \dfrac{M}{2} < y \le M, 1 \le x \le M \end{cases}$$

$$f_3(x,y) = r_5,\ 1 \le x \le M,\ 1 \le y \le M$$

In the expressions above, $r_i,\ i = 1, \ldots, 5$ represents a constant parameter.

Consider now a population of random vectors of the form:

$$\underline{g}(x,y) = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ g_3(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.

(i) Find the images $g_1(x,y)$, $g_2(x,y)$ and $g_3(x,y)$ using the Karhunen-Loeve (KL) transform. **[6]**
(ii) Comment on whether you could obtain the result of (b)-(i) above using intuition rather than by explicit calculation. **[2]**

2. (a) (i) Perform histogram equalization on the 8×8 image shown in **Figure 1** below, where the pixels have integer values from the set $\{0,1,\ldots,7\}$. Clearly state any assumptions you make. **[3]**

(ii) Provide the histogram of the equalised image. **[3]**

(iii) Plot the transformation function that has yielded the equalized image. **[3]**

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 4 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 4 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 4 \\ 1 & 1 & 1 & 2 & 2 & 2 & 3 & 5 \\ 2 & 2 & 2 & 2 & 2 & 5 & 5 & 5 \\ 6 & 6 & 6 & 6 & 7 & 7 & 5 & 5 \end{bmatrix}$$

Figure 1

(b) Explain the effects of performing histogram equalization on an image that has already undergone the process. In this question the image intensity is assumed to be a continuous random variable with a known probability density function. **[3]**

(c) Consider the spatial filter H given by

$$H = \begin{bmatrix} -1 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

and assume a 6-bit grayscale image I.

(i) Determine the maximum and minimum possible values that a pixel of image I, to which this spatial filter is applied, can have. Do not apply any type of normalization. **[4]**

(ii) In order to ensure that the output of this filter remains a 6-bit grayscale image, we need to propose a suitable gray level transformation function to the intensities of the filtered image. Find a transformation function that would achieve this goal. **[4]**

3. We are given the degraded version g of an image f such that in lexicographic ordering

$$g = Hf + n \quad (1)$$

where H is the degradation matrix which is assumed to be block-circulant and n is the noise term which is assumed to be zero-mean, white and independent of the image f. All images involved have size $N \times N$ after extension and zero-padding.

(a) In a particular scenario the degradation process can be modelled as a linear filter with the two-dimensional impulse response given below:

$$h(x,y) = \begin{cases} 0.125 & x = 0, y = \pm 1 \\ 0.75 & x = 0, y = 0 \\ 0 & \text{elsewhere} \end{cases}$$

Investigate whether there are frequency pair(s) for which Inverse Filtering cannot be applied. **[6]**

(b) As an archaeologist, I possess a collection of 40 ancient manuscripts that are similar in size and appearance. Unfortunately, half of these manuscripts have been permanently misplaced. However, I had the foresight to acquire digital pictures of all 40 manuscripts using the same camera, lighting conditions, position, and distance parameters. Regrettably, all the pictures appear blurry due to a camera misfocusing issue during acquisition. Fortunately, I can offer you access to the original camera and comprehensive information about the acquisition conditions, allowing you to recreate the precise conditions under which the pictures were taken. Your task is to help me restore the images to their original quality, enabling me to continue my research.

(i) Provide a method to obtain the degradation model which caused the blurring. Use the fact that many blurred images are available, in order to remove the additive camera noise which cannot be assumed to be negligible. **[7]**

(ii) Explain how you can use Wiener filtering to deblur the lost images, making use of the information and camera I provided and your knowledge of image processing. Assume that all available images have the same power spectrum. **[7]**

Assume that the degradation process for this problem can be modelled as in Equation (1) above.

4. (a) Consider an image whose pixel values range from 0 to 7. The probability of any pixel having a value of 0 is 0.51 and the remaining pixel values all have equal probability.

(i) Find the entropy of this image in bits per pixel. **[3]**

(ii) Construct a Huffman code for the gray levels assuming that each of the pixel values are coded separately. **[3]**

(iii) Find the average number of bits per pixel of the coded set of values. **[3]**

(b) For a source with q symbols, find the maximum possible value of the entropy. **[3]**

(c) A digital image of size 100×100 pixels has five distinct intensity levels. We aim to code this image in a lossless manner.

(i) What is the minimum number of bits required if no additional information is available? **[2]**

(ii) What is the minimum number of bits required if the intensity levels are distributed as shown in **Table 1** below? **[3]**

intensity	**Number of pixels**
r_1	2000
r_2	4000
r_3	1000
r_4	?
r_5	2000

Table 1

(iii) For each of the above two cases, construct an optimal code assuming each pixel value is coded individually. **[3]**